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Commission errors of active intentions: the roles of aging, cognitive load, and practice

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Performing an intended action when it needs to be withheld, for example, when temporarily prescribed medication is incompatible with the other medication, is referred to as commission errors of prospective memory (PM). While recent research indicates that older adults are especially prone to commission errors for finished intentions, there is a lack of research on the effects of aging on commission errors for still active intentions. The present research investigates conditions which might contribute to older adults' propensity to perform planned intentions under inappropriate conditions. Specifically, disproportionally higher rates of commission errors for still active intentions were observed in older than in younger adults with both salient (Experiment 1) and non-salient (Experiment 2) target cues. Practicing the PM task in Experiment 2, however, helped execution of the intended action in terms of higher PM performance at faster ongoing-task response times but did not increase the rate of commission errors. The results have important implications for the understanding of older adults' PM commission errors and the processes involved in these errors.

Keywords: prospective memory; commission errors; cue saliency; automatic processes; intention retrieval practice

The ability to remember to fulfill an intention at a certain point in the future is referred to as prospective memory (PM). While most studies in this domain focus on the ability to execute an intended action in the appropriate moment, recent studies also indicate a surge of interest in commission errors in PM (Bugg & Scullin, 2013; Pink & Dodson, 2013; Scullin & Bugg, 2013; Scullin, Bugg, & McDaniel, 2012; Scullin, Bugg, McDaniel, & Einstein, 2011; Scullin, Einstein, & McDaniel, 2009; Walser, Fischer, & Goschke, 2012). Commission errors refer to the execution of an intended action such as taking medication after dinner when, in fact, the intended action should not be performed (Scullin & Bugg, 2013), for example, because prescription has changed. Examples such as this highlight the importance of being able to deactivate intentions when the planned intention is no longer relevant. Previous research indicates that older adults might be especially susceptible to such commission errors (Scullin et al., 2012). So far, however, the study of commission errors in PM was limited to *deactivated* (i.e., finished or cancelled) intentions (but see Walser et al., 2012; West & Craik, 1999). Everyday life, on the other hand, often affords situations in which an intention may not be finished or cancelled as a whole but should nonetheless be withheld under certain conditions. For example, the intention to take blood thinners should be withheld on the day of surgery. The present research thus

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aims to fill this gap by focusing on factors that may increase older adults' risk of executing *active* delayed intentions in inappropriate situations.

Commission errors in prospective memory

In laboratory PM tasks, participants are first asked to form an intention, usually to press a certain key (PM key) in response to a target (PM cue) and then perform an ongoing task in which PM cues are embedded (Einstein & McDaniel, 1990). This paradigm mimics the situation of having to remember a previously formed intention while being engaged in some other activity. In order to study commission errors of deactivated intentions in this paradigm (e.g., Scullin & Bugg, 2013; Scullin et al., 2011, 2012), participants first perform an ongoing task with embedded PM cues. In the next phase, participants are instructed that the intention to press the PM key is finished and thus no longer relevant (i.e., intention deactivation) and they perform another task, again with embedded PM lures (i.e., stimuli that served as PM cues in the previous phase). Scullin et al. (2011) showed that the slowing of response times of older adults to PM cues compared with control stimuli during the finished phase was substantially stronger than in younger adults. This slowing effect was interpreted as indicating involuntary retrieval of the previous intention which in turn points to failures of intention deactivation. That is, older adults were less likely than younger adults to effectively deactivate the intention when it was no longer relevant. Importantly, in this study participants did not perform any commission errors. Commission errors (i.e., pressing the PM key in the finished phase) occurred, however, when the background in the PM phase and in the finished phase were identical and the PM cues in both phases were salient (Scullin & Bugg, 2013; Scullin et al., 2012). This effect was larger in older than in younger adults, again suggesting that older adults are specifically impaired in deactivating no longer relevant intentions (Scullin et al., 2012). Furthermore, strengthening the association between PM cue and intended action via an implementation intention encoding strategy, which is assumed to increase automatic processing of PM cues (e.g., McDaniel & Scullin, 2010; McFarland & Glisky, 2012; Rummel, Einstein, & Rampey, 2012), increased commission errors in older and younger adults (Bugg, Scullin, & McDaniel, 2013), a result pointing to the presumably automatic nature of commission errors.

In a similar paradigm, Pink and Dodson (2013) had participants perform two blocks of a lexical decision task with embedded PM cues. Cues for the two blocks differed, but the second block also contained the PM cues from the first block, and participants were specifically instructed to only respond to the cues from the present block with the PM key. Participants who performed an additional digit monitoring task during the second block showed more commission errors in response to these PM lures than participants without additional task load but only when the PM response was practiced in the first block. However, source monitoring errors may contribute to commission errors in this paradigm: Because participants need to remember whether the cue belongs to the no longer relevant cues from the first block or to the now relevant cues of the second block, failures of source monitoring may have exacerbated the rate of commission errors.

Nonetheless, results such as these are taken as indication that commission errors occur rather automatically and put special demands on control mechanisms to withhold PM responses (Scullin & Bugg, 2013). Cognitive aging in general has been shown to be associated with declines in various cognitive functions (Salthouse, 1988), one of them being inhibitory processes (Hasher, Stolzhus, Zacks, & Rypma, 1991). According to this view, older adults suffer from an inhibitory deficit that manifests itself in, among others, greater difficulties in inhibiting inappropriate responses. In line with this notion, measures

of cognitive control were shown to be predictive of commission errors in older adults (Scullin et al., 2012). Automatized cognitive processes, on the other hand, are assumed to be largely spared from aging (e.g., Schmitter-Edgecombe, 1999). Accordingly, older adults might be equally likely to rely on automatic processes for PM responding as younger adults (Einstein, McDaniel, & Scullin, 2011; McDaniel & Einstein, 2011), and at the same time, they might have more difficulties in inhibiting automatically activated PM responses in inappropriate contexts.

In a related paradigm, West and Craik (1999) investigated age differences in commission errors to similar but not identical PM lures. In this study, participants had to respond to green uppercase words with a special PM key press, and words that partially matched these criteria (i.e., either gray uppercase words or green lowercase words) served as lures for the PM intention. While younger adults produced virtually no commission errors to partially matched PM lures, older adults did (see also West & Covell, 2001, for a similar finding). Furthermore, there is evidence from studies using event-related potentials that the presentation of partially matched PM lures as well as of PM lures after intention deactivation elicit an early activation pattern similar to the one associated with the presentation of the PM cues themselves (West, 2011; West, McNerney, & Travers, 2007). While the activation pattern of partially matched PM lures seems to occur consistently in younger adults, it has been shown to be less consistent in older adults (West & Covell, 2001). The authors interpret these findings as evidence that the ability to distinguish between PM cues from lures is reduced with age. Thus, commission errors in the partial-match paradigm are likely due to other processes than commission errors in other paradigms.

In line with Meiser and Rummel (2012), we argue that PM lures not only are noticed but also trigger the automatic initiation of the intended action. Thus, inhibitory control is required to prevent the execution of intended actions in response to lures and, due to the age-related inhibitory deficit (Hasher et al., 1991), older adults can be assumed to have special difficulties in doing so. Because previous studies mostly focused on to-be-deactivated intentions, it is unclear whether automatic action initiation or failures of intention deactivation drove the increased rate of commission errors in older adults. Although the partial-match paradigm used by West and Craik (1999) avoids this problem, older adult's commission errors in this paradigm could have been due to their impaired ability to distinguish cues from partial-match cues.

The present research

We adopted the paradigm introduced by Meiser and Rummel (2012) to extend the study of commission errors to still active intentions. In this paradigm, the intention to press the PM key is associated with the occurrence of members of a certain word category (e.g., animal names) in an ongoing lexical decision task. All stimuli in the lexical decision task vary randomly along a context dimension such as font color, and participants are instructed to only press the PM key when members of the target category occur in a certain context such as animal names in green font color (i.e., the “execute” context). Animal names that are presented in yellow font (i.e., the “suppress” context), on the other hand, are to be considered regular ongoing task stimuli, rendering PM key presses to target words in the wrong font color commission errors. This paradigm extends the finished-phase paradigm by presenting participants with PM cues (“execute” context) and PM lures (“suppress” context) within the same task phase.

Furthermore, this paradigm differs subtly but in important ways from the paradigm employed by West and Craik (1999). Whereas PM lures in West and Craik's study

matched PM cues only partially, all animal names in the present paradigm *are* PM cues but given a certain context feature, the PM response should be withheld. Thus, because in the present paradigm all animal names fulfill the criterion of PM cues, PM performance is not a matter of distinguishing cues from non-cues as in the West and Craik study but a matter of inhibiting (automatically) activated PM responses to cues. Consequently, withholding inappropriate PM responses in the present paradigm should be more difficult than in the West and Craik paradigm. In fact, Meiser and Rummel (2012) found commission errors in younger adults, whereas West and Craik (1999) did not. Furthermore, imagery instructions, which are thought to automatize PM responding, increased the rate of commission errors in Meiser and Rummel (2012). This result further strengthens the interpretation of commission errors in the present paradigm as tapping inhibitory processes for withholding the PM response to cues in the wrong context. Therefore, the paradigm from Meiser and Rummel seems to be optimally suited to investigate inhibitory demands induced by PM lures for active intentions.

Previous research has shown that increasing the degree to which participants rely on automatic PM processes increased the rate of commission errors in this paradigm (Meiser & Rummel, 2012; Experiment 2), a result in line with the interpretation of commission errors as arising from largely automatic processes (e.g., Meiser & Rummel, 2012; Scullin & Bugg, 2013). Furthermore, many studies suggest that automatic memory processes (Schmitter-Edgecombe, 1999) and processes of prospective remembering in particular (Kliegel, Jäger, & Phillips, 2008; Mullet et al., 2013; Scullin et al., 2011; Wang et al., 2011) may be spared by aging. In line with the interpretation of commission errors as resulting from failures of cognitive control, automatically activated PM actions in the “suppress” context should require cognitive effort to be withheld. Therefore, we argue that older adults will produce more commission errors in the “suppress” context than younger adults. As an alternative hypothesis, commission errors might not differ between older and younger adults in this paradigm because older adults are less likely to notice PM cues while performing an ongoing task as compared with younger adults (Henry, Rendell, Phillips, Dunlop, & Kliegel, 2012). Thus, older adults’ PM impairment could, in fact, decrease their risk of commission errors for active intentions because if they are less likely to notice PM cues (and lures) they are also less likely to respond to them.

We conducted two experiments in which we extended the study of commission errors in older adults to still active intentions. In the first experiment, we compared older and younger adults on PM performance for salient PM targets and imposed an additional cognitive load in a group of younger adults in order to investigate the role of executive resources for inhibiting automatically activated intentions. Assuming that practice of the PM task would increase reliance on automatic processes, we compared PM hits and commission errors in older adults and younger adults between a practiced PM task and a PM task without prior practice in the second experiment.

Experiment 1

The goal of Experiment 1 was to test the hypothesis that older adult’s impaired efficiency of cognitive control processes increases their rate of commission errors for active intentions as compared with younger adults. To provide an even more stringent test of this hypothesis, we also included another group of younger participants with a secondary tone-monitoring task in addition to the ongoing task to reduce their cognitive processing capacity. If commission errors are driven by largely automatic processes (Meiser & Rummel, 2012; Scullin et al., 2009) and suppressing them draws on inhibition resources, then reduced cognitive

processing capacity should reduce the ability to inhibit the initiated action and thus increase the rate of commission errors even in younger adults. In line with this assumption, previous research showed that divided attention increases the risk of repetition errors (McDaniel, Bugg, Ramuschkat, Kliegel, & Einstein, 2009) and commission errors in the case of finished intentions in younger adults (Pink & Dodson, 2013).

To facilitate cue noticing, we employed salient cues in both the “suppress” and in the “execute” context. In an earlier study in the paradigm of finished intentions, rates of commission errors were highest when the cues were salient, but almost absent when the cues were not salient (Scullin et al., 2012). Furthermore, it has been shown that noticing of PM cues requires less attention when they are salient in the context of the ongoing task (e.g., Einstein, McDaniel, Manzi, Cochran, & Baker, 2000; Marsh, Hicks, & Hancock, 2000; Rummel, Boywitt, & Meiser, 2011).

To summarize, reduced cognitive control resources (due to aging or due to cognitive load) are assumed to increase the probability of commission errors due to failures of inhibiting inappropriate PM responses in the wrong context. This hypothesis builds on the assumption that PM cues are noticed as such in both contexts and that the PM response in the “suppress” context needs to be inhibited in order to avoid a commission error. To test this assumption, we also analyzed response times to correctly rejected PM lures (lure interference) and expected these response times to be specifically slowed due to the controlled inhibition engaged upon lure occurrence.

Method

Participants and design

Seventy-three younger adults (58 female, mean age = 21.86; $SD = 2.15$; range = 18–29, mean years of schooling = 13.12) were randomly assigned to the full attention condition or the cognitive load condition and received eight euros as compensation. The 39 older adults (24 female; mean age = 72.54 years; $SD = 4.69$; range = 66–81, mean years of schooling = 12.20) were recruited from a pool of healthy community-dwelling adults who had already participated in earlier studies at the psychology department. The older adults were screened for a history of neurological diseases and received 12 euros as compensation. All participants were native speakers of German, and none was colorblind.

The context of to be “suppressed” or “executed” PM responses was varied within subjects and will be described in the next section.

Procedure and materials

The experiment started with instructions for a lexical decision task, which emphasized both speed and accuracy. For this task, 186 words and 12 animal names of medium frequency and of similar length (between 5 and 10 letters) were randomly selected from a German word database (CELEX; Baayen, Piepenbrock, & Gulikers, 1995). Nonwords were constructed by randomly swapping two letters in half of the words. At the beginning of each lexical decision trial, a fixation cross of variable duration (250–750 ms) was presented in the center of the screen followed by a stimulus (i.e., a word, a nonword, or an animal name) presented in capital letters. Stimuli remained on the screen until a response was entered. Participants were asked to press the “A” key for words and the “L” key for nonwords and were subsequently given six practice trials half of which were in yellow font color and the other half in green font color. Then, participants were given the opportunity to ask questions.

The younger participants in the cognitive load condition were additionally informed about an added tone-monitoring task. For the tone-monitoring task, participants were presented with a random sequence of two tones differing in pitch (440 and 330 Hz) during the lexical decision task. Each tone lasted for 1 s followed by an inter-stimulus interval of 2 s. Participants were instructed to press the space bar whenever three consecutive tones of the same pitch were presented during the lexical decision task. Only the younger adults in the cognitive load condition heard tones during the lexical decision task.

Then, for all participants the prospective intention was induced by asking them to respond to animal names in the lexical decision task by pressing the “7” key. In order to make sure they would find the key, the participants were then asked to locate and press the “7” key. There were no instructions regarding the order of responses on PM cue trials. But participants were informed that PM key presses after word/nonword key presses would be also counted as correct responses. In order to render PM cues and PM lures salient, all animal names but none of the other stimuli in the lexical decision task were enclosed with arrows (“>>animal name<<”). Participants were informed about this in the instructions (see Marsh et al., 2000, Experiment 3, for a similar salience manipulation).

The 12 animal names for the PM task were the German words for “fox,” “hamster,” “dragonfly,” “carp,” “zebra,” “llama,” “lizard,” “horse,” “wasp,” “badger,” “pike,” and “sparrow.” Half of the animal names were presented in the “suppress” context color, and the other half were presented in the “execute” context color with both contexts appearing in a fixed randomized order. Animal names in the “execute” context were always presented on trials 16, 48, 112, 128, 160, and 192, and animal names in the “suppress” context were always presented on trials 32, 64, 80, 96, 144, and 176. The animal names were randomly assigned to the trials with an “execute” and “suppress” context for each participant anew. Assignment of colors (green vs. yellow) to the “suppress” vs. “execute” context was counterbalanced across participants.

Then, all participants were informed to only press the “7” key in response to animal names in green (yellow) font but not to press the “7” key in response to animal names in yellow (green) font. After the experimenter confirmed that participants understood all instructions, a 3-min visual-search task commenced. Then, the 192 trials of the lexical decision task started without any further mentioning of the PM task. Half of the stimuli of each type were presented in green font, and the other half were presented in yellow font. After the lexical decision task, participants’ memory for the PM task instructions as well as for the intention-relevant font color was probed, and participants completed a paper and pencil demographic questionnaire. Last, participants were debriefed and dismissed.

Results

We excluded data from all participants who both did not remember the PM task at the end of the experiment and did not press the PM key at least once during the lexical decision task. Also, data from all participants who did not remember the correct font color for the PM response were excluded to rule out commission errors due to confusion of the context colors. Applying these criteria led to the final cell sizes of $n = 32$ younger adults in the full attention condition, $n = 33$ younger adults in the cognitive load condition, and $n = 33$ older adults.

Ongoing task performance

Ongoing task performance was assessed via response times and accuracy. Table 1 displays mean accuracy and reaction time data.

Table 1. Means (*SEs*) of ongoing task performance.

	Accuracy (%)		Response times (ms)	
	Execute context	Suppress context	Execute context	Suppress context
<i>Experiment 1</i>				
Younger adults full attention	95.51 (1.20)	95.80 (1.00)	807 (45)	805 (50)
Younger adults cognitive load	92.67 (1.20)	92.38 (0.90)	955 (45)	955 (50)
Older adults	94.00 (1.20)	95.68 (0.90)	1256 (45)	1287 (49)
<i>Experiment 2</i>				
Younger adults control	95.08 (0.89)	96.23 (0.50)	867 (37)	884 (43)
Younger adults practice	95.31 (0.60)	95.35 (0.55)	811 (23)	817 (25)
Older adults control	96.51 (1.27)	96.23 (1.73)	1326 (61)	1298 (62)
Older adults practice	91.36 (3.74)	92.35 (3.68)	1172 (48)	1167 (47)

Accuracy in the ongoing task was computed as the proportion of correct responses to words and nonwords. A mixed analysis of variance (ANOVA) on accuracy rates in the ongoing task with the between-participants factor group (younger adults full attention, younger adults cognitive load, older adults) and the repeated measures factor context (execute, suppress) yielded a marginally significant main effect of age group, $F(2, 95) = 2.76$, $p = .068$, $\eta_p^2 = .055$. No other main effect or interaction reached significance, both F s < 1.2 . Post-hoc *LSD* comparisons revealed that the accuracy rates for the younger adults under cognitive load ($M = 92.50$; $SE = 1.00$) were significantly lower than for younger adults under full attention ($M = 95.70$; $SE = 1.00$), $p = .03$, and marginally lower than for older adults ($M = 94.80$; $SE = 1.00$), $p = .10$. Accuracy rates for younger adults under full attention and for older adults did not differ significantly, $p = .56$.

Response time analyses were confined to words because PM cues and PM lures were always words and thus word response times should better reflect PM processes (Brewer, 2011). Only accurate responses and responses slower than 300 ms but faster than 2 *SDs* above each participant's individual mean were included in the analysis to control for aberrant response times (Ratcliff, 1993).

A 3×2 mixed ANOVA with the between participants factor group (younger adults full attention, young adults cognitive load, older adults) and the repeated measures factor context (execute, suppress) on mean response times yielded a main effect of group, $F(2, 95) = 26.13$, $p < .001$, $\eta_p^2 = .355$. No other main effect or interaction yielded significance, both $F < 1$. *LSD* comparisons revealed that the response times of the younger adults under full attention ($M = 806$; $SE = 47$) were significantly faster than those under cognitive load ($M = 955$; $SE = 46$), $p = .026$. Both younger age groups were still significantly faster in the ongoing task than the older adults ($M = 1,271$; $SE = 46$), both $p < .001$.

Prospective memory performance

The proportions of PM responses in both contexts (i.e., pressing the “7” key in response to an animal name in the “execute” or the “suppress” context) were compared to evaluate PM performance. Any presses of the “7” key occurring at any time between the presentation of an animal name and during the presentation of the next ongoing task stimulus were scored as a PM response. PM responses on words or nonwords other than animal names occurred in less than 0.7% of all ongoing task trials and similarly often in the three groups

Table 2. Means (*SEs*) of prospective memory performance and interference.

	PM responses		
	Execute context	Suppress context	Proportion change score
<i>Experiment 1</i>			
Younger adults full attention	.97 (.01)	.08 (.02)	2.13 (.21)
Younger adults cognitive load	.95 (.02)	.19 (.04)	1.83 (.10)
Older adults	.86 (.03)	.21 (.05)	1.79 (.10)
<i>Experiment 2</i>			
Younger adults control	.79 (.05)	.03 (.02)	1.18 (.03)
Younger adults practice	.91 (.03)	.08 (.03)	1.21 (.04)
Older adults control	.72 (.05)	.15 (.05)	1.16 (.05)
Older adults practice	.87 (.04)	.13 (.05)	1.27 (.05)

Notes: PM = prospective memory. The proportion change score is computed as the mean reaction time to correctly rejected PM targets in the “suppress” context divided by the mean reaction time to correct word trials in the ongoing task and should be indicative of the degree of lure interference.

and were thus not further considered. Mean proportions of PM responses are displayed in Table 2.

A mixed ANOVA with the between-participants factor group (younger adults full attention, younger adults cognitive load, older adults) and the repeated measures factor context (execute, suppress) on PM responses yielded a significant main effect of context, $F(1, 95) = 810.53$, $p < .001$, $\eta_p^2 = .90$. The main effect of group was not significant, $F(2, 95) = 1.15$, $p = .320$, $\eta_p^2 = .02$, but the main effect of context was qualified by a significant interaction with group, $F(2, 95) = 6.52$, $p = .002$, $\eta_p^2 = .12$. Pair-wise *LSD* comparisons on the rate of PM responses in the execute context revealed that older adults produced lower PM hit rates than younger adults in the full attention condition, $p = .002$, and than younger adults in the cognitive load condition, $p = .006$. However, the two groups of younger adults did not differ significantly from each other, $p = .680$.

PM responses in the suppress context, on the other hand, were more frequent in the older adults than in the younger adults in the full attention condition, $p = .031$. Similarly, commission errors were marginally more frequent in younger adults under cognitive load than in younger adults under full attention, $p = .056$. Younger adults with cognitive load and older adults did not differ significantly in terms of PM responses in the suppress context, $p = .796$. Thus, younger adults in the cognitive load condition produced similar rates of commission errors as the older adults, but their PM hit rates were similar to hit rates of the younger adults without cognitive load.

Lure interference

Response times to correctly classified PM cues in the “suppress” context were analyzed to investigate differences in slowing due to inhibiting PM responses in the wrong context. Accordingly, response times on trials with lexical decision responses in the “suppress” context should be slower than response times on regular ongoing task trials. To account for the generally prolonged response times of the older adults as compared with younger adults, the mean latencies of lexical decision responses to animal names in the “suppress” context were divided by the mean response times for correctly classified words in the “suppress” context. The resulting score indicates the proportional change in response

times from ongoing task trials to PM trials in the “suppress” context, with scores larger than 1 indicating slowing and scores lower than 1 indicating speed-up on PM lure trials. It is important to note that this proportional slowing score avoids finding artifactual interactions between age group and trial type due to differences in baseline performance (i.e., removable interactions; Wagenmakers, Krypotos, Criss, & Iverson, 2012). Mean interference scores are displayed in Table 2.

The slowing effect was significant (i.e., larger than 1) in all groups, all $ps < .001$. A one-way ANOVA on this proportional change score for PM lures with the between-subjects factor group (young adults cognitive load, younger adults full attention, older adults) did not reveal any significant group differences, $F(2, 91) = 1.59$, $p = .210$, $\eta_p^2 = .034$, suggesting proportional levels of slowing due to rejecting PM lures across all groups.

Discussion

The central finding of the first experiment is the mirror effect (Glanzer & Adams, 1990) like pattern of lower rates of correct PM responses accompanied by higher rates of commission errors in older as compared with younger adults in the full attention condition. Younger adults with cognitive load performed similarly as the older adults in terms of commission errors but produced similar levels of correct PM responses as the younger adults without cognitive load. The older adults’ reduced rate of successful PM responses in the “execute” context relative to the younger adults suggests that older adults were less likely than younger adults to detect the PM cues (and thus probably also the PM lures). The increased proportion of commission errors in older adults, however, indicates that older adults’ inhibitory control was impaired, so that the initiation of the intended action could not always be successfully suppressed once a cue was detected. These results are well in line with the hypothesis that reduced cognitive control resources (due to aging or due to cognitive load) increase the probability of commission errors due to failures of inhibiting inappropriate PM responses in the wrong context.

The substantial slowing of response times in lexical decisions for animal names in the “suppress” context in older and younger adults can be taken as an indication of automatic intention retrieval requiring inhibitory control to prevent commission errors to lures (Scullin et al., 2012). The lack of age differences in slowing on PM lure trials suggests that older adults did not exhibit the typically found greater difficulties in task coordination than younger adults in the present paradigm (cf. Ihle, Hering, Mahy, Bisiacchi, & Kliegel, 2013).

The conclusion emerging from the first experiment is that even when failures of intention deactivation play no role, older adults are still more likely to perform commission errors than younger adults in response to salient PM lures. This is especially remarkable given the lower correct PM response rate of older adults as compared with younger adults. Thus, even though PM cues were less likely to be noticed by older adults than by younger adults they elicited commission errors more frequently in older than in younger adults. In Experiment 2, we went one step further and tested whether non-salient PM cues also elicit commission errors of active intentions.

Experiment 2

In most PM studies and in many everyday situations, PM cues are not particularly salient. Therefore, it is important to investigate whether an increased risk of commission errors in older compared to younger adults still occurs when PM cues are not salient.

We did not include a group with cognitive load in this study, but we manipulated PM task practice between participants. Many important prospective intentions in everyday life resemble rather habitual intentions because they are not only performed once but repeatedly, such as always taking medication after dinner. Nonetheless, there are situations in which such intended actions need to be withheld, for example, on the day of one's surgery. If practice of a PM task reduces the reliance on controlled processes in remembering to perform the intention, then practice should increase the rate of commission errors. Previous research suggests that older adults (or younger adults under cognitive load) are more likely to incorrectly repeat PM responses for practiced PM tasks than for non-practiced tasks (Einstein, McDaniel, Smith, & Shaw, 1998; McDaniel et al., 2009; Pink & Dodson, 2013). Thus, practice of the PM task could be expected to increase commission errors in older but not in younger adults because older adults are assumed to have greater difficulties in inhibiting automatically activated response tendencies than younger adults.

Method

Participants and design

Seventy-seven younger adults (59 female; mean age = 22.17 years; $SD = 2.62$; range = 18–30; mean years of schooling = 13.29) participated and received eight euros as compensation. The 79 older participants (53 female; mean age = 72.56 years; $SD = 4.52$; range = 65–84; mean years of schooling = 12.18) were sampled from the same pool of healthy community-dwelling adults as in Experiment 1 and received 10 euros for their participation. All participants were native speakers of German and none was colorblind. Participants were randomly assigned to the practice and control condition. As in Experiment 1, context was manipulated within subjects.

Procedure and materials

The procedure and materials were largely identical to Experiment 1 with the following exceptions. After the six practice trials for a lexical decision task, participants in the practice condition were given the PM task to respond to animal names in the lexical decision task by pressing the “7” key. As in Experiment 1, participants were then asked to locate and press the “7” key. Then, all participants completed a block of 48 trials of the lexical decision task with 12 PM cues embedded. PM cues for the practice phase were the 12 animal words used in Experiment 1. This time, however, PM cues were not enclosed by arrows (i.e., were not salient). Half of all words, half of all nonwords, and half of all PM cues were presented in green font. The other half of all stimuli were presented in yellow font. Importantly, only the participants in the practice condition were instructed to respond to the animal names by pressing “7”. For the other participants this task was yet another practice block of the lexical decision task.

After the practice block, the PM intention was also induced in the participants in the no-practice condition via the same instructions as in the practice condition. Next, all participants were informed to only press the “7” key in response to animal names in green (yellow) font but not to press the “7” key in response to animal names in yellow (green) font. After the experimenter confirmed that participants understood all instructions, a 3-min visual-search task commenced to delay the PM intention. Then, the 192 trials of the lexical decision task started without any further mentioning of the PM task. The 12 animal

names from the practice phase occurred again and without arrows on trials 16, 48, 112, 128, 160, and 192 (“execute”) and on trials 32, 64, 80, 96, 144, and 176 (“suppress”).

Results

Applying the same exclusion criteria as in Experiment 1 lead to the final sample of $n = 34$ younger adults in the no-practice condition, $n = 40$ younger adults in the practice condition, $n = 33$ older adults in the no-practice condition, and $n = 35$ older adults in the practice condition.

Ongoing task performance

Accuracy and response times in the lexical decision task were computed as in Experiment 1 and mean accuracy rates and response times are displayed in Table 1.

A mixed ANOVA with the between-participants factors age group (older vs. younger) and PM condition (practice vs. no-practice) and the within-participants factor context (suppress vs. execute) was conducted with accuracy rates in the lexical decision task as dependent variable. Of the main effects only context approached significance, $F(1, 138) = 3.43$, $p = .066$, $\eta_p^2 = .024$, all other $p > .47$. This main effect was qualified by the significant three-way interaction, $F(1, 138) = 5.26$, $p = .023$, $\eta_p^2 = .037$. Simple main effect analyses indicated that younger adults in the no-practice condition showed higher accuracy for words in the “suppress” context than in the “execute” context, $F(1, 138) = 5.52$, $p = .020$, $\eta_p^2 = .038$, all other $ps > .082$, a result with no obvious relevance for the present research goal.

Mean response times for correct lexical decision responses to words were submitted to the same analysis revealing a main effect of age group, $F(1, 135) = 83.48$, $p < .001$, $\eta_p^2 = .382$, with substantially slower response times in older than in younger adults. The effect of condition was also significant, $F(1, 135) = 5.52$, $p = .02$, $\eta_p^2 = .039$, reflecting faster responding in the practice condition than in the no-practice condition. Of all the other effects, only the interaction between context and age group was marginally significant, $F(1, 135) = 3.74$, $p = .055$, $\eta_p^2 = .027$, all other $ps > .249$. Simple main effect analyses revealed that for older adults in the control condition response times were marginally slower in the execute than in the suppress context, $F(1, 135) = 3.52$, $p = .063$, $\eta_p^2 = .025$ while there was no difference between the two contexts for any other factor combination, all $ps > .257$.

Prospective memory performance

PM responses to words that were not animal names occurred in less than 0.1% of all ongoing task trials and similarly often in the two age groups. They were thus not further considered. Mean proportions of PM responses are displayed in Table 2.

In the first step, we compared the rate of “7” key presses to PM cues in the practice block between older and younger adults in the practice condition but no differences emerged, $t(71) = 0.68$, $p = .502$, $d = 0.16$. Thus, older and younger adults performed equally well in the practice block.

Next, a mixed ANOVA with the between-participants factors age group (younger adults vs. older adults) and condition (practice vs. no-practice) and the repeated measures factor context (execute vs. suppress) on PM responses in the experimental block was conducted. This analysis yielded a significant main effect of context, $F(1, 138) = 465.17$,

$p < .001$, $\eta_p^2 = .771$, indicating higher rates of PM responses in the “execute” context than in the “suppress” context. The effect of practice condition was also significant, $F(1, 138) = 11.37$, $p = .001$, $\eta_p^2 = .076$, with higher rates of PM responses in the practice condition than in the control condition. The interaction between context and age group was marginally significant, $F(1, 138) = 3.89$, $p = .051$, $\eta_p^2 = .027$, as was the interaction between context and practice condition, $F(1, 138) = 3.22$, $p = .075$, $\eta_p^2 = .023$. However, neither the main effect of age group nor the interaction between age group and practice or the three-way interaction were significant, all F s < 1 . Simple main effect analyses indicated that practice selectively increased PM responses in the “execute” context, $F(1, 138) = 10.18$, $p = .002$, $\eta_p^2 = .069$, but not in the “suppress” context, $F < 1$. Further, simple main effect analyses revealed that older and younger adults differed in the rate of commission errors, $F(1, 138) = 4.49$, $p = .036$, $\eta_p^2 = .031$, but not with regard to correct PM responses $F(1, 138) = 1.53$, $p = .219$, $\eta_p^2 = .011$.

Lure interference

As in Experiment 1, we computed a proportional change score for latencies of correct lexical decision responses to animal names in the “suppress” context (see Table 2). This change score was significantly larger than 1 in all groups, indicating significant slowing, all p s $< .001$. An ANOVA with the between participants factors age group (younger adults vs. older adults) and condition (practice vs. no-practice) on these interference scores yielded a marginally significant main effect of practice, $F(1, 125) = 2.99$, $p = .086$, $\eta_p^2 = .023$, with higher interference scores in the practice condition than in the control condition. No other effect was statistically significant, all p s $> .40$.

Discussion

The results of Experiment 2 replicate the central finding of Experiment 1. Older adults were more likely to produce commission errors in response to lures than younger adults even though the PM cues and lures were not salient this time. The results further show that PM practice substantially increased PM performance in both older and younger adults, indicating that the manipulation was successful. The significant decrease in ongoing task response times of both older and younger adults in the practice condition as compared with the no-practice condition further suggests that fewer resources were necessary to achieve higher PM performance after practice. This finding is in line with previous research showing that PM practice can improve PM (McDaniel & Scullin, 2010). PM practice, however, did not increase commission error rates for younger or older adults, which could have been expected based on the assumption that practice automatizes PM responding. Indeed, Pink and Dodson (2013) who also employed a practice manipulation found increased rates of commission errors after practice when young adults were set under cognitive load. One possible reason why practice did not increase commission errors in our group of older adults might be that the practice in the study by Pink and Dodson was more extensive (i.e., responding 10 times to each cue). However, as PM performance in the present study was improved after practice, our practice manipulation changed PM processing to some extent nonetheless. We will further discuss this result in the General Discussion.

As in Experiment 1, we observed substantially slowed responses to correctly rejected lures. Notably, both older and younger adults showed the same proportion of slowing but the slowing was only marginally higher after PM practice. Inasmuch as response times to

correctly rejected lures are a measure of the resources engaged to inhibit inappropriate PM responses to lures, this result provides tentative evidence that inhibitory demands may increase with PM practice. However, further research is necessary to substantiate this marginal effect. Nonetheless, in line with Experiment 1, the present pattern of results suggests that lure inhibition puts similar demands on older and younger adults but that older adults are more likely to fail in inhibiting the PM response in the inappropriate context, even when PM cues are not particularly salient.

General discussion

Commission errors of prospective remembering may have serious consequences in everyday life. Whereas previous research focused on age differences in commission errors for finished or suspended intentions, the present research extends this line of research to age differences in PM commission errors of active intentions. In this regard, the present research offers a variety of novel results. First, both experiments revealed higher rates of commission errors in older as compared with younger adults. Importantly, that was the case with both salient (Experiment 1) and non-salient (Experiment 2) cues. It was hypothesized that older adults might not be more prone to commission errors than younger adults in the case of non-salient cues because they are *a priori* less likely to notice the PM cues and thus have fewer opportunities to commit such errors. The increased proportions of commission errors in older adults across both experiments suggest, however, that older adults noticed some of the PM lures and had then difficulties inhibiting automatically initiated intention execution, although context information signaled it was not appropriate.

In addition to PM hits and commission errors, we considered proportional changes in lexical decision response times to correctly classified PM lures in the “suppress” context as a measure of inhibitory demands. In both experiments interference from lures was similar for older and younger adults. Thus, inhibiting PM responses to lures seems to be similarly demanding for both age groups. However, as evident from the increased commission error rates, older adults are nevertheless less likely to actually succeed in inhibiting the PM response to lures.

Furthermore, the practice manipulation in Experiment 2 provides novel evidence regarding the processes underlying commission errors. In line with previous research (McDaniel et al., 2009), practice of the PM task improved PM. This was evident from the increased rate of correct PM responses in older as well as younger adults. Notably, the increase in PM performance was accompanied by a speed-up in ongoing task response times in both age groups. Thus, in combination, these results support the assumption that practice renders PM responding more automatic. However, practice did not increase the rate of commission errors and increased the level of lure-induced interference only marginally. This latter effect was comparably large in both age groups. Thus, PM practice came at a cost of lure-induced slowing but not of an actual increase in commission errors. Therefore, it seems that inhibition of PM responses to lures in the inappropriate context was especially demanding after PM practice.

In sum, the results not only add to the evidence showing that older adults might be more prone to commission errors than younger adults but also allow important insights into the processes giving rise to commission errors. We discuss potential accounts of commission errors and how the present data are informative for these accounts in the following section.

Explaining commission errors

In general, commission errors may be explained via different mechanisms: Either as arising due to controlled monitoring processes or as arising due to rather automatic processes. While both accounts agree that failures to inhibit the activated PM response contribute to commission errors, the accounts differ in their explanation of how PM responses are activated by PM cues. According to accounts which assume that retrieval of delayed intentions can be achieved via automatic processes (McDaniel & Einstein, 2007), commission errors to PM lures are due to spontaneous noticing processes triggered by PM-relevant cues which bring to mind the intention to press the PM key, even when these targets occur in an inappropriate context. Temporary failures of cognitive control may then lead to the execution of the intended action despite the wrong context. According to this interpretation, the higher rate of commission errors observed in older adults as well as in younger adults under cognitive load in Experiment 1 can be explained by preserved automatic processes (Mullet et al., 2013) in combination with a (load- or age-induced) lower efficiency in cognitive control mechanisms (Hasher et al., 1991; West, 1996). This interpretation is in line with the suggestion of Meiser and Rummel (2012) that commission errors to PM cues in the wrong context are indicative of automatic processes of PM. The positive effect of practice on PM hit rates in combination with faster response times in the ongoing task as well as marginally increased interference due to lures in Experiment 2 suggest that practice automatized PM performance to some degree. The null effect of practice on rates of commission errors, on the other hand, is difficult to reconcile with this interpretation.

Alternatively, commission errors may be explained by accounts which assume that resource-demanding monitoring processes are always necessary for retrieving an intended action (e.g., Smith, 2003). According to such accounts, identification of both PM cues and lures must be due to controlled processes and decreased cognitive resources in terms of working memory capacity, for example (Smith & Bayen, 2005), should therefore reduce the ability to retrieve the intention. Given that older adults are assumed to exhibit reduced cognitive resource, the pattern of decreased PM performance in older adults in Experiment 1 is in line with this account. In this view, however, lower rates of PM performance should have been also observed in younger adults under cognitive load and should generally be accompanied by lower rates of commission errors because reduced cognitive resources should be associated with reduced rates of overall intention retrieval independent of the context in which a cue occurs. Of course, scarce cognitive resources are also detrimental to inhibiting false responses to lures, but this aspect only becomes relevant once lures are detected and elicit a PM response. Older adults' reduced rates of PM performance accompanied by increased rates of commission errors in the present experiments are therefore difficult to reconcile with the assumption that controlled monitoring processes (in combination with failures of inhibition) drive commission errors.

There is at least one other potential explanation for commission errors of active intentions which does not resort to failures of inhibition: The task of remembering to perform an intended action requires joint retrieval of the intended action and the appropriate circumstances (i.e., the correct context) at the right moment. In this view, either controlled or automatic processes might retrieve the intention to press the PM key, but if retrieval of the intention is not followed by retrieval of the correct context for its execution, then a commission error occurs. First, it is important to note that we excluded all participants from further analysis who were not able to distinguish the "execute" from

the “suppress” context after the lexical decision task. Second, the data speak against this alternative explanation: If failures of retrieving the intention and relevant context jointly yielded commission errors, increased rates of intention retrieval due to practice (Experiment 2) should be accompanied by increased rates of commission errors. Put more generally, higher rates of PM hits should be accompanied by higher rates of commission errors because the more often the intention is retrieved the more often failures of context retrieval can come into play. However, increased rates of PM performance without increased rates of commission errors after practice in Experiment 2 speak against this explanation. Furthermore, substantial negative correlations between PM performance and commission errors in older adults ($r = -.441$, $p < .001$) and in younger adults ($r = -.294$, $p = .011$) are further evidence against this account because they suggest that higher rates of correct PM performance were, in fact, associated with lower rates of commission errors. Thus commission errors cannot be solely due to failures to retrieve the correct context at the right moment.

To summarize, most pieces of evidence in the present study are in line with the interpretation of commission errors arising due to automatic PM processes in combination with failures of action inhibition (Meiser & Rummel, 2012; Scullin et al., 2011, 2009) except for one result: commission errors did not increase as a function of PM practice. This result appears to be puzzling but could be reconciled with the automaticity account of commission errors by assuming that participants in the practice condition had more resources available for inhibiting automatically activated PM responses due to the facilitation by practice than participants in the control condition. This assumption is supported by the speed-up of response times in combination with increases of PM hits in the practice condition as compared with the control condition.

Conclusions

The present findings show that older adults are more prone than younger adults to commission errors in inappropriate contexts while the intention is still active, thus extending the empirical basis of age-related effects on PM commission errors. This appears to be true even when older adults are less likely to notice the PM cues and to respond to them in the right moment. In sum, converging evidence points to the role of preserved automatic processes and reduced inhibitory control in older age for this mirror-effect like pattern of PM performance.

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